Aurora v1.2 Cycle-Accurate Simulator Specification

AUR-SIM-001 | Canonical Draft 1

1. Simulator Modes

- Functional mode: architectural correctness without cycle timing
- Cycle-accurate mode: pipeline, window, scoreboard, memory and interrupt timing
- Trace-replay mode: deterministic reproduction of failures

2. State Model

- GPRs, PSR, A0/A1, Q0/Q1, PC/SP/LR
- Four-entry issue window and per-entry status
- Scoreboard and outstanding completion records
- Memory map, TCM/SRAM timing, MMIO and DMA state
- Interrupt pending/active state and countdown timer

3. Per-Cycle Order

1. Apply external events and memory responses
2. Retire oldest completed instructions
3. Recognize precise interrupts
4. Advance multi-cycle units
5. Execute/launch selected issue pair
6. Select and compact issue window
7. Fetch and decode new words

4. Trace Format

- Cycle number, PC, decoded instruction, issue lane, resource mask
- Register/flag/accumulator writes
- Memory requests/responses and stream consumption
- Window contents, scoreboard stalls and branch recovery
- Interrupt and exception events

5. Validation

The simulator must consume the canonical ISA JSON and compare against directed tests, random programs, and RTL traces.